Power: USB-C, U_IN, charger, 5 V buck, 3V3

power.kicad_sch

Battery pack, protection, gauge

battery.kicad_sch

MCU, UI, LCD, switches

mcu.kicad_sch

I2C mux, expander, 5 V peripheral ports

ports.kicad_sch

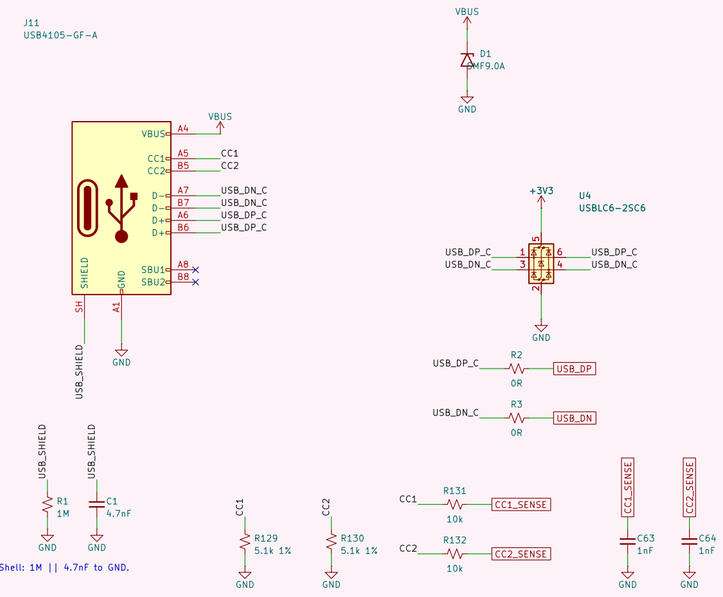
Spec docs/SPEC.md v1.0; generated by build/scripts/gen_sch.py
5 V USB-C (no PD), U_IN eFuse, BQ25887 2S charger, TPS56637 2S -> 5 V buck, MAX17205, 5 x 5

Sheet: /
File: E0005_controller.kicad_sch

Title: E0005 ESP32-S3 5 V controller

Size: A4	Date: 2026-10-27	**Rev: A**
KiCad E.D.A. 10.99.0-unknown		Id: 1/5

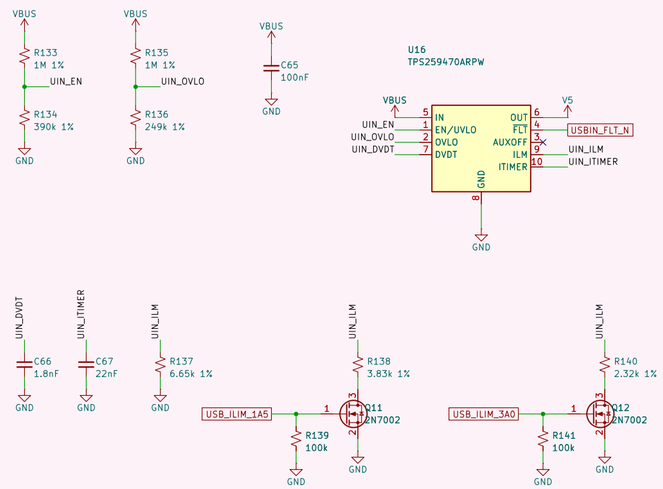
USB-C J11 (5 V, no PD), VBUS TVS, USB ESD, CC Rd + sense



Shell: 1M || 4.7nF to GND.

D1 SMF9.0A: 15.4 V clamp < 20 V BQ25887 / TPS25947 abs max. USBLC6 VBUS pin to +3V3.
D+/D− 90 ohm diff., OR to GPIO20/19. CC: 5.1k Rd each (dead-battery safe):
10k + 1 nF to GPIO6/7 (ADC1): -0.41 V default, -0.92 V 1.5 A, -1.68 V 3.0 A.

USB INPUT eFUSE U_IN TPS259470A: VBUS -> V5, 0.50 / 1.37 / 2.81 A



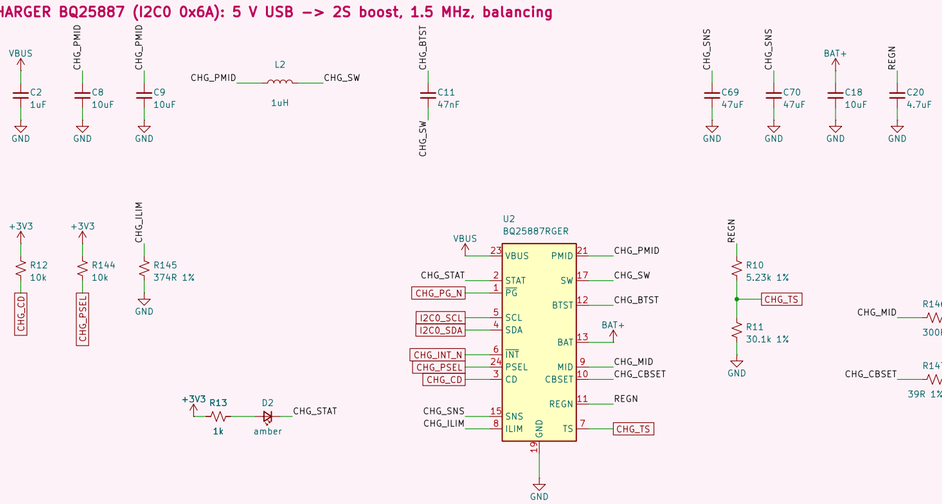
I_LIM = 3334 / R_ILM: 6.65k -> 0.50 A (reset), || 3.83k -> 1.37 A (P06), || 2.32k -> 2.81 A (P06+P07).
Legs are cumulative: never P07 without P06. 100k gate pull-downs -> 0.50 A at reset.
OVLO 1M/249k: 6.02 V rising. EN/UVLO 1M/390k: 4.28 V rising / 3.88 V falling.
ITIMER 22 nF (~18 ms). dVdt 1.8 nF. FLT -> TCA9535 P15 (low = at its limit / TSD).
Thermal: keep the SPEC 5.6.1 copper (IN/OUT pours >= 80 mm2, 6 vias each, no reliefs).

V5 SENSE (GPIO4), POWER FLAGS



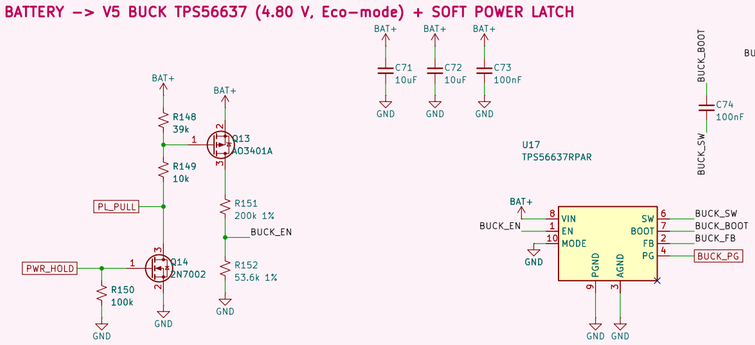
V5 / 2 -> GPIO4 (ADC1_CH3). 2.75 V at 5.5 V.
V5 and BAT+ are driven by U16 OUT / U2 BAT.

CHARGER BQ25887 (I2C0 0x6A): 5 V USB -> 2S boost, 1.5 MHz, balancing



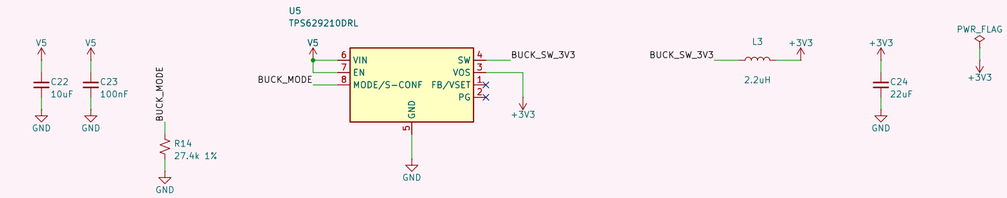
Per BQ25887 Fig. 69: VBUS 1 uF, PMID 2 x 10 uF/25 V, L 1 uH PMID -> SW, BTST 47 nF, SNS 2 x 47 uF/16 V,
BAT 10 uF, REGN 4,7 uF, TS: REGN 5.23k / 30.1k || NTC1 103AT-2 (battery sheet) -> JEITA 0-60 C.
ILIM 374R -> 2.97 A hardware cap (K_ILIM 1110). PSEL 10k up -> 500 mA default input limit.
CD 10k up -> HIZ / charging OFF at reset (TCA9535 P05 drives low). PG -> P16, INT -> GPIO5.
MID 300R (bottom-cell reverse protection). CBSET 39R 2512 -> ~108 mA balancing (0.45 W).
BAT = BAT+ after F1. SNS is the boost output (internal R_SNS to BAT): caps only.

BATTERY -> V5 BUCK TPS56637 (4.80 V, Eco-mode) + SOFT POWER LATCH



FB 69.8k/10k: 0.6 x (1 + 6.98) = 4.79 V. FF 20k + 100 pF (Table 4, 5 V row). MODE = GND -> Eco-mode
(no reverse current). Table 4 5 V row: L1 3.3 uH; V5 total ~88 uF effective (2 x 47 uF + 6 x 10 uF). <= 100 uF.
EN 200k / 53.6k from Q_PL: 5.38 V rising (5.76 V max), 4.30 V falling (Ip 1 uA, Ih 4 uA): 1.99 V at 8.4 V.
Latch: Q_PL AO3401A (-30 V, +-12 V), 39k gate-source -> VGS -6.7 V at 8.4 V. Press: PB_RAW low -> D13
(MCU sheet) pulls PL_PULL -> Q_PL on -> buck on -> firmware sets PWR_HOLD (GPIO36). Off: PWR_HOLD low.
USB + battery: keep PWR_HOLD high (EN low turns on the 200R SW discharge, ~25 mA from V5).
PG -> TCA9535 P17 (10k up on ports sheet).

3V3 BUCK TPS629210 (from V5, 1 A)



MODE/S-CONF 27.4k: VSET mode, auto PFM/PWM, output discharge. FB/VSET open -> 3.3 V.
EN tied to V5; 100 % duty down to ~3.5 V V5. PG unused.

Spec docs/SPEC.md v1.0; generated by build/scripts/gen_sch.py
5 V USB-C (no PD), U_IN eFuse, BQ25887 2S charger, TPS56637 2S -> 5 V buck, MAX17205, 5 x 5

Sheet: /Power: USB-C, U_IN, charger, 5 V buck, 3V3/
File: power.kicad_sch

Title: E0005 ESP32-S3 5 V controller

Size: User	Date: 2026-10-27	Rev: A
KiCad E.D.A. 10.99.0-unknown		Id: 2/5

CELLS (bottom side), F1, REVERSE-CELL LEDS, NTCs

PWR_FLAG
PWR_FLAG
PWR_FLAG
B_N
S8252_VDD
GAUGE_VBATT

CELL2_P
F1
BAT+
10A T

CELL2_P
BH2
Keystone 1121 + P50B
Cell 2 (top)
MID

MID
R15
2.2k
D3
BAT54W
D4
red
CELL2_P

MID
BH1
Keystone 1121 + P50B
Cell 1 (bottom)
B_N

B_N
R16
2.2k
D5
BAT54W
D6
red
MID

Reverse-cell LED: R + BAT54W + red LED, oriented to light only when that cell is reversed.

CHG_TS
RT1
10k B3435
GND

GAUGE_AIN1
RT2
10k B3435
GND

RT1 (NTC1) -> BQ25887 TS, RT2 (NTC2) -> gauge AIN1; both between the cells.

PROTECTION S-8252AAG + LOW-SIDE FETS, R_SENSE

CELL2_P
R17
470R
S8252_VDD

MID
R18
470R
S8252_VC

S8252_VDD
C25
100nF
S8252_VC

S8252_VC
C26
100nF
B_N

U6
S-8252AAG-M6T1U
S8252_VDD 5 VDD
S8252_VC 4 VC
B_N 6 VSS
DO 1 DSG_G
CO 2 CHG_G
VM 3 S8252_VM

S8252_VM
R19
2k
GND

S8252_VDD
D7
BAT54W
S8252_VC

S8252_VC
D8
BAT54W
B_N

FET_D
Q1
PSMN2R0-30YLDX
DSG_G 4 G
D 5
S 1
B_N

FET_D
Q2
PSMN2R0-30YLDX
CHG_G 4 G
D 5
S 1
SNS_P

SNS_P
R20
5mR 1%
GND

Pack path: Cell1(−) = B_N -> Q1 (DSG, S at B_N) -> FET_D -> Q2 (CHG) -> SNS_P -> R_SENSE 5 mOhm -> GND.
S-8252AAG: OV 4.30 V, UV 2.80 V, OC 0.150 V (VM after R_SENSE -> -17-21 A, short guard): F1 = OC limit.
R 470R + C 100nF per ABLIC Table 11; 2k on VM. BAT54W clamps: reverse-cell protection of VC.
Cell OV ladder: VREG 4.20 V -> S-8252 4.30 V -> BQ25887 cell OVP 4.37 V. No BQ29209 (owner Q2).

FUEL GAUGE MAX17205 (stack-only, board GND)

BAT+
R28
49.9R
GAUGE_VBATT

GAUGE_VBATT
C31
100nF
GND

U15
MAX17205G+
GAUGE_VBATT 5 VBATT
7 CELL1
6 CELL2
13 CELLX
GND
SNS_GND
GAUGE_CSN 4 CSN
2 CSP
SDA 12 GAUGE_SDA
SCL 11 GAUGE_SCL
ALRT1 8 GAUGE_ALRT
THRM 14 GAUGE_THRM
AIN1 9 GAUGE_AIN1
AIN2 10
GND
REG3 1 GAUGE_REG3
REG2 3 GAUGE_REG2
GND 15
GND

NT1
GAUGE_CSN
GND

GAUGE_SDA
R29
150R
I2C0_SDA

GAUGE_SCL
R30
150R
I2C0_SCL

GAUGE_ALRT
R31
1k
GAUGE_ALRT_N

GAUGE_THRM
R32
10k 1%
GAUGE_AIN1

GAUGE_REG3
C32
0.47uF
GND

GAUGE_REG2
C33
0.47uF
GND

Per MAX17205 Fig. 2 (low-side protector): GND/EP = board GND, CELL1/2/X = GND, PackCfg BtEn=1, CxEn=0, ChEn=0.
CSP = SNS_P (FET side of R_SENSE), CSN via NT1 Kelvin tap. VBATT 49.9R + 100 nF from BAT+.
THRM -> 10k -> AIN1 + RT2. AIN2 unused (GND). I2C0 0x36/0x0B, 150R series; ALRT 1k -> GPIO35.

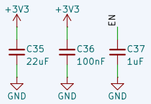
Spec docs/SPEC.md v1.0; generated by build/scripts/gen_sch.py
5 V USB-C (no PD), U_IN eFuse, BQ25887 2S charger, TPS56637 2S -> 5 V buck, MAX17205, 5 x 5

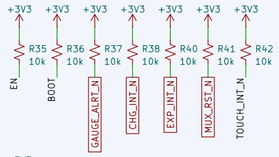
Sheet: /Battery pack, protection, gauge/
File: battery.kicad_sch

Title: E0005 ESP32-S3 5 V controller

Size: User	Date: 2026-10-27	Rev: A
KiCad E.D.A. 10.99.0-unknown		Id: 3/5

ESP32-S3-WROOM-1-N16R2

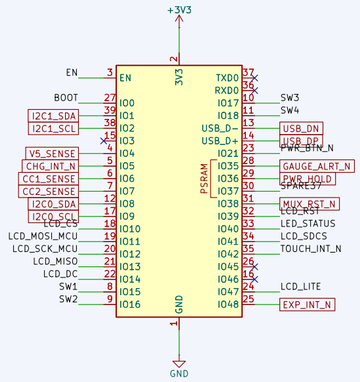








U1
ESP32-S3-WROOM-1-N16R2

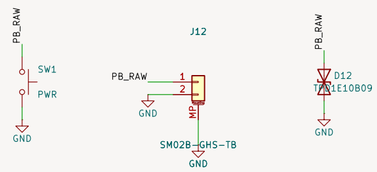


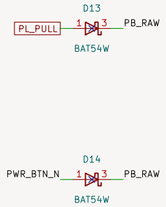




EN 10k / 1 uF RC. Strapping: GPIO0 BOOT 10k up; GPIO3, 45, 46 NC. No test points (owner rule).
N16R2: quad PSRAM, GPIO35–37 usable (35 gauge ALRT, 36 PWR_HOLD, 37 spare).
ADC1: GPIO4 V5_SENSE, GPIO6/7 CC1/CC2_SENSE (power sheet). GPIO48 = TCA9535 INT.
I2C0 (system) 2.2k pull-ups here. Interrupt lines 10k pull-ups.

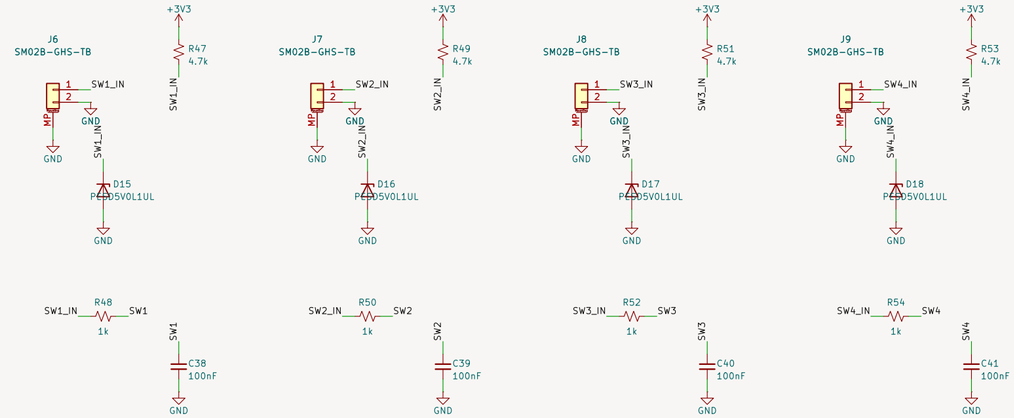
POWER BUTTON SW1 + J12 (soft power latch on, GPIO21 sense)





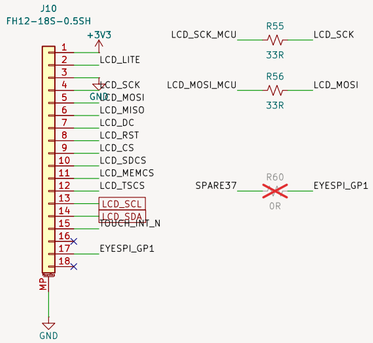
Press: PB_RAW low -> PL_PULL low (latch on, power sheet) + GPIO21 low (wake). Long press: firmware drops PWR_HOLD. PB_RAW floats up to BAT+ when idle: 9 V ESD (not PESD5V0), GPIO21 diode blocks it.

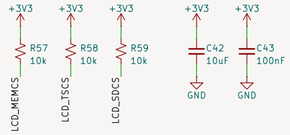
SWITCH INPUTS J6–J9 (JST-GH 2p: 1 = SWn, 2 = GND)



4.7k pull-up at the connector, ESD, 1k series, 100 nF at the GPIO (GPIO15–18, RTC wake).

LCD J10: EYESPI 18p 0.5 mm FPC (Adafruit 4311 / 2.8" EYESPI)





Pin order from Adafruit EYESPI breakout (EYE_SPI_DISPLAY_BOTCONTACT): 1 VIN 2 LITE 3 GND 4 SCK
5 MOSI 6 MISO 7 DC 8 RST 9 TCS 10 SDCS 11 MEMCS 12 TSCS 13 SCL 14 SDA 15 INT 16 BUSY 17 GP1 18 GP2.
SCL/SDA = mux ch5 (4.7k on ports sheet). Unused selects 10k up. GP1 via DNP 0R from GPIO37; GP2 NC.

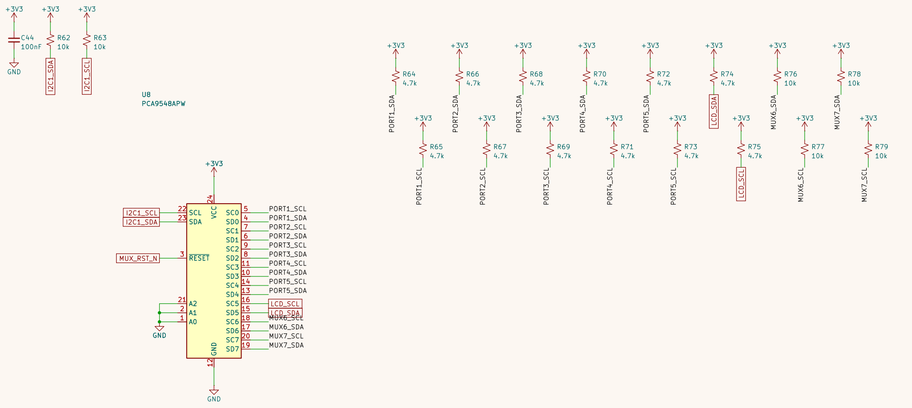
Spec docs/SPEC.md v1.0; generated by build/scripts/gen_sch.py
5 V USB-C (no PD), U_IN eFuse, BQ25887 2S charger, TPS56637 2S -> 5 V buck, MAX17205, 5 x 5

Sheet: /MCU, UI, LCD, switches/
File: mcu.kicad_sch

Title: E0005 ESP32-S3 5 V controller

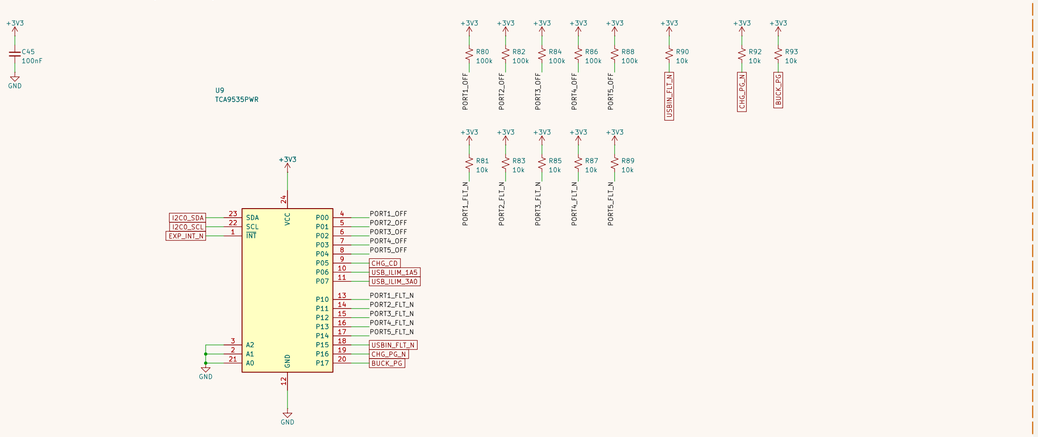
Size: User	Date: 2026-10-27	**Rev: A**
KiCad E.D.A. 10.99.0-unknown		Id: 4/5

I2C1 MUX PCA9548APW (0x70)



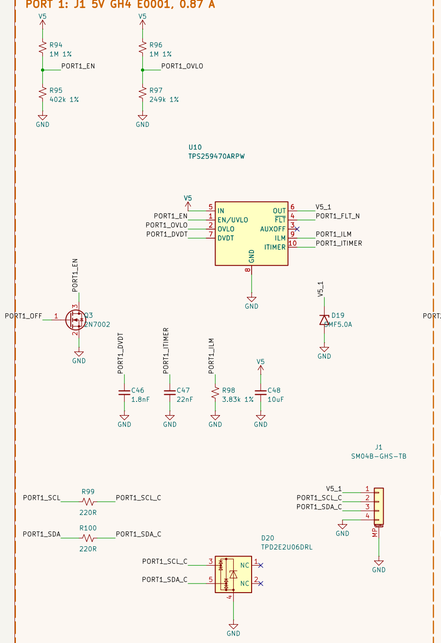
A0–A2 = GND → 0x70. RESET ← GPIO38 (10k up on MCU sheet). Upstream 10k pull-ups.
ch0–ch3 → J1–J4 (E0001 5 V rev.), ch4 → J5 (E0002 5 V rev.), ch5 → EYESPI touch: 4.7k to 3V3.
ch6/ch7 spare: 10k pull-ups. Firmware: ≤ 4 channels open, writes only.

SYSTEM I/O EXPANDER TCA9535 (I2C0 0x20)



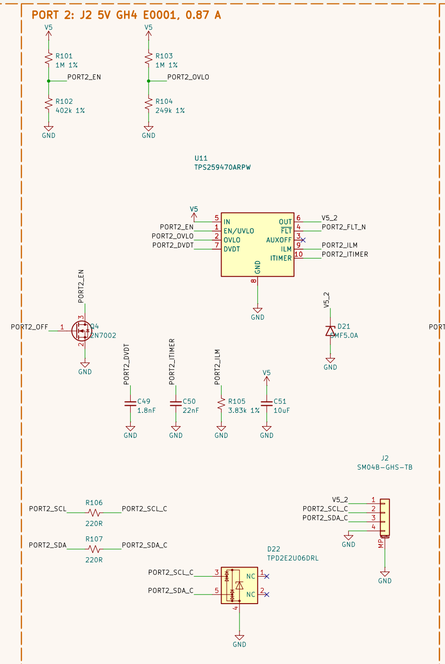
Reset state (all inputs): P00–P04 PORTn_OFF 100k up → ports OFF; P05 CHG_CD 10k up (power sheet)
→ charging OFF; P06/P07 USB_ILIM_1A5/3A0 100k down at the leg FETs → U_IN 0.50 A.
P10–P14 port FLT; P15 USBIN_FLT_N, P16 CHG_PG_N, P17 BUCK_PG: inputs, 10k pull-ups. INT → GPIO48.

PORT 1: J1 5V GH4 E0001, 0.87 A



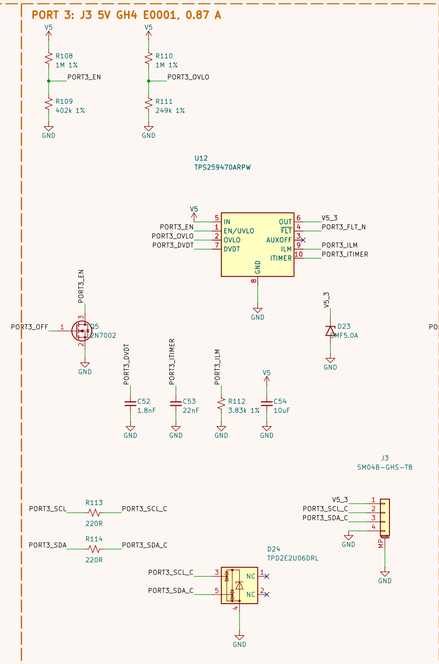
Silkscreen: "5V" + "1" + "E0001"
I_LIM 0.87 A (0.74–1.00); OVLO 6.02/5.47 V; UVLO 4.19/3.80 V

PORT 2: J2 5V GH4 E0001, 0.87 A



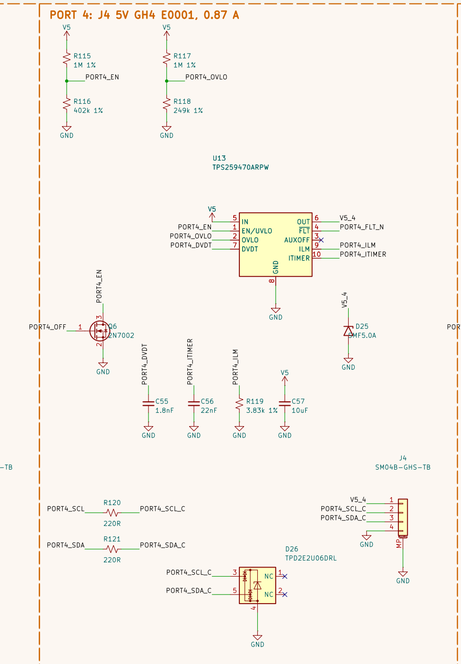
Silkscreen: "5V" + "2" + "E0001"
I_LIM 0.87 A (0.74–1.00); OVLO 6.02/5.47 V; UVLO 4.19/3.80 V

PORT 3: J3 5V GH4 E0001, 0.87 A



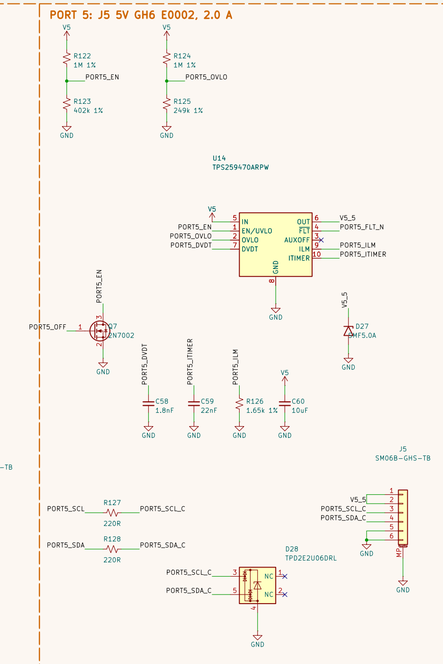
Silkscreen: "5V" + "3" + "E0001"
I_LIM 0.87 A (0.74–1.00); OVLO 6.02/5.47 V; UVLO 4.19/3.80 V

PORT 4: J4 5V GH4 E0001, 0.87 A



Silkscreen: "5V" + "4" + "E0001"
I_LIM 0.87 A (0.74–1.00); OVLO 6.02/5.47 V; UVLO 4.19/3.80 V

PORT 5: J5 5V GH6 E0002, 2.0 A



Silkscreen: "5V" + "5" + "E0002"
I_LIM 2.03 A (1.80–2.20); OVLO 6.02/5.47 V; UVLO 4.19/3.80 V

Spec docs/SPEC.md v1.0; generated by build/scripts/gen_sch.py
5 V USB-C (no PD), U_IN efuse, BQ25887 2S charger, TPS56637 2S → 5 V buck, MAX17205, 5 × 5

Sheet: /I2C mux, expander, 5 V peripheral ports/
File: ports.kicad_sch

Title: E0005 ESP32-S3 5 V controller

Size: User	Date: 2026-10-27	Rev: A
KiCad E.D.A. 10.99.0-unknown		Id: 5/5